



ZALLIANT

Lightweight Heat Stress Detection for Cow Health Monitoring

Meet The Team



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Introduction

Improving cattle health starts with detecting issues early, before they become too costly

Zalliant

Saving farmers money by transforming cattle herd management from **reactive** to **proactive**

- Combines proprietary sensor technology with data processing to help farmers make informed treatment decisions
- Promotes cattle health by detecting illnesses early, reducing antibiotic use and healthcare costs



An image of Zalliant's bolus. The device is swallowed by the cow and remains in the rumen (stomach) for the lifetime of the device

The Stakes: Why Cattle Health Monitoring Matters

22%

Proportion of agriculture sector dominated by cattle

\$600-900

Economic loss per cow from reduced milk output

\$444

Average mastitis cost per cow in the first 30 days of lactation

THE CORE PROBLEM



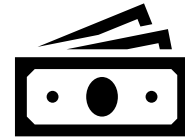
Traditional cattle health monitoring relies on manual observation



Farmer walks into the field and looks for signs of visible distress



By the time symptoms are visible, the illness has progressed significantly



Treatment costs soar, productivity drops, and mortality risk spikes.

Problem Statement and Goals

Our goal: Evaluate the economic viability of implementing ML-based methods into Zalliant's workflow.

Is it possible to create a model that will:

- Improve or match current performance of AWS lambda functions' capability to temperature event detection
- Operate at a lower cost than current methods

Our Approach

Explore various machine learning algorithms and benchmark against Zalliant's current methods as well as original alternative statistical approaches.

How can we reduce costs?

- Create lightweight algorithms, reducing computational costs
- Create robust algorithms that function at lower sampling rates, extending the lifetime of each device

Outline

Project Timeline & Milestones

Data Overview

Baseline Reconstruction

Detection Model

Model Evaluation Results

Achievements & Economic Impact

Challenges & Next Steps

Acknowledgements

Project Timeline & Milestones



Project Timeline & Milestones

Data Acquisition & Project Kick Off

Obtained Access
to Data
Dashboard and
Kicked Off Project

Blue Line Reconstruction

Reconstructed
the Zalliant blue
line via
approximation

Model Evaluation

Validated against
current alerts,
demonstrating
impressive metrics
and earlier detection

EDA & Visualization

Conducted
EDA on cow
types, outliers
and data gaps

Model Exploration

Developed a novel,
custom statistical
model tailored to
Zalliant's use case

Data Overview



Raw Data: Rumen Temperature Data

- 20 cows – 10 from each farm, with and without alerts
- 30-day span
- Time series has gaps as sensor data may not always be capturable



Data Source: CowMonitor.com

- Grace-Way Farms (New York, USA)
- Voltaurin (South France)

< Animal Id : <1453>

Sensor Serial Number: 231600000NY, 25114000BKV Pen: 1 Age: 5 years DOB: 10-01-2020 Health Index Count: 0

2026-04-22

7 Days

Refresh Data

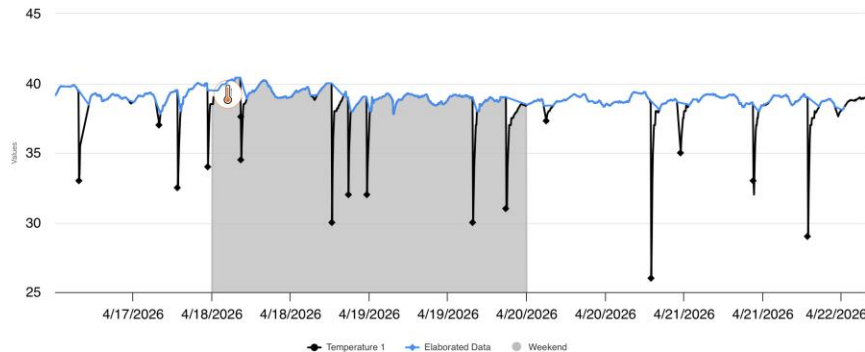
Export
Data

Export Elaborated
Data

| Latest Data: 03:52 2026-04-22

C

F



Time-series data as shown in Zalliant's proprietary dashboard

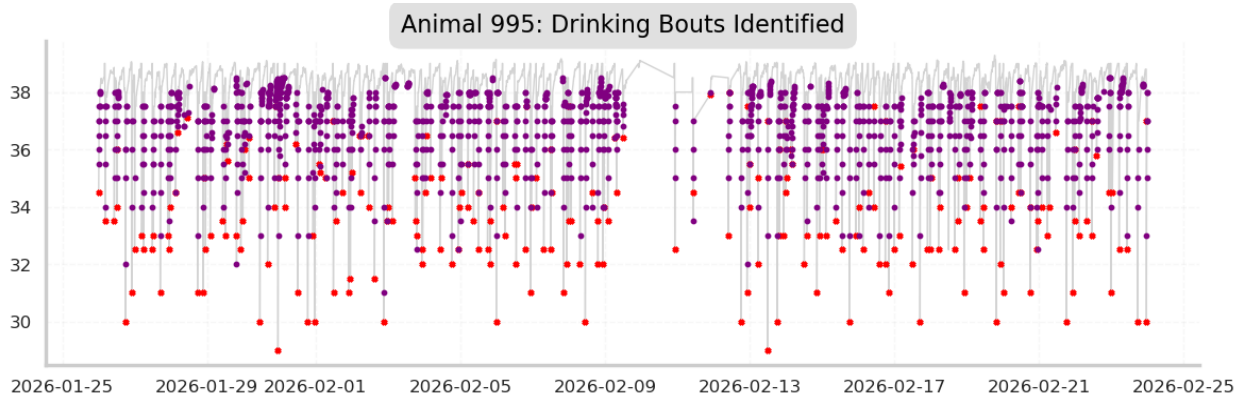
Raw Dataset Variables

Variable	Description
animal_id	Unique identifier assigned to each cow (convention varies by farm)
temperature (°C)	Primary signal: rumen temperature captured by bolus sensor, updated every 5 minutes
device_id	Unique identifier for each bolus device (a cow may have multiple over time due to battery replacement, etc.)
timestamp (UTC)	Date and time associated with each observation
alert	Boolean flag triggered when the current detection system identifies a potential health event
communication_flag	Boolean flag indicating connectivity issues between sensor and network

Blue Line in Stages

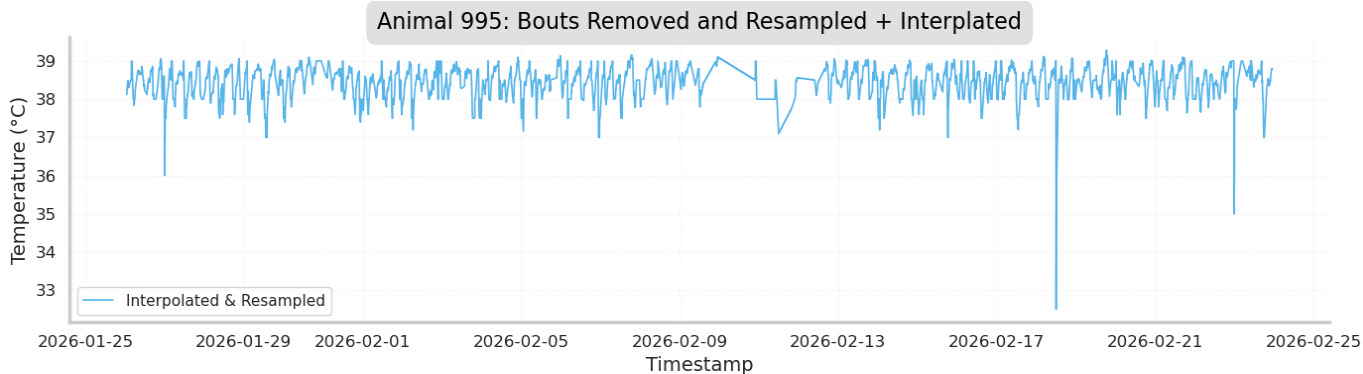
Stage 1: Identify

temperatures
corresponding
to drinking
bouts using
water alerts.



Stage 2:

Remove bouts
from the data
with linear
interpolation.



Establishing Blue Line

Challenge: Temperatures can only be estimated to be associated with a bout.

Fixed Threshold Method

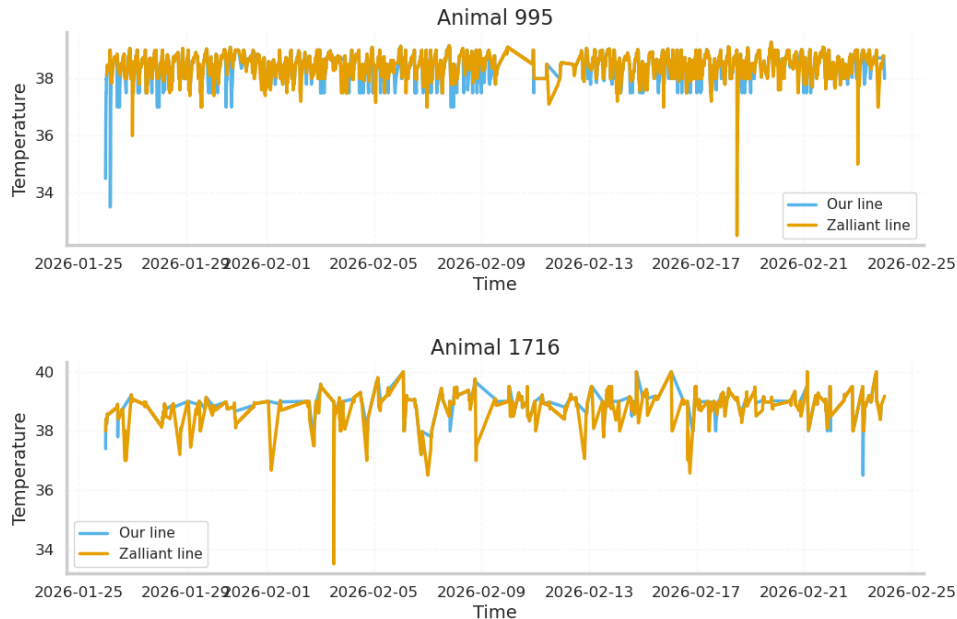
- Detects drinking events when temperature drops $\geq 2^{\circ}\text{C}$ between readings

Rolling Threshold Method

- Uses recent temperature behavior (rolling mode and standard deviation) to detect abnormal drops
- Closest approximation to Zalliant's line built with proprietary algorithms.

Global Threshold

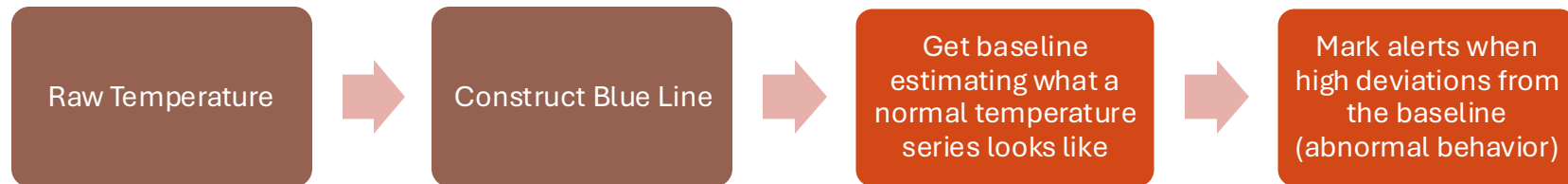
- Temperatures below 37.5°C are treated as outliers and removed before construction



From Baseline Reconstruction to Predictive Modeling

After reconstructing the baseline temperature, we then:

- estimate each animal's normal temperature trajectory
- measure deviations from baseline instead of relying on raw readings
- use those deviation signals for alert forecasting



Models tested but not used:

- XGBoost,
- LSTM (Long Short Term Memory)
- Logistic Regression

Models Tested

Logistic Regression

- Classified alert events using engineered temporal features (lags, rolling statistics, temp. changes)
- Not chosen: It requires labeled data (limited ground truth), performances varied across cows, sensitive to class imbalance

XGBoost

- Predicted short term predictive signals (ROC AUC: 0.7, precision: 0.52, recall: 0.53)
- Not chosen: Predictions were too noisy and unstable, not reliable for real-time event triggering, dependent on limited ground truth labels

LSTM

- Modeled temperature as a time sequence, learned patterns, and used past readings to predict future alert events
- Not chosen: Moderate and inconsistent performance ($F1 = .53$), computationally expensive and slow

Key Takeaway: Performance depended heavily on ground truth labels, and these ML-based models lacked stability for reliable event detection

Our Method: Context Aware Event Detection

Current Zalliant's Alerts

- Based on quarterly rolling averages and fixed thresholds
- Highly sensitive to sensor noise, data gaps, and short-term behaviors (e.g., drinking)
- Flags isolated points, not sustained patterns of abnormality

Our Proposed Method

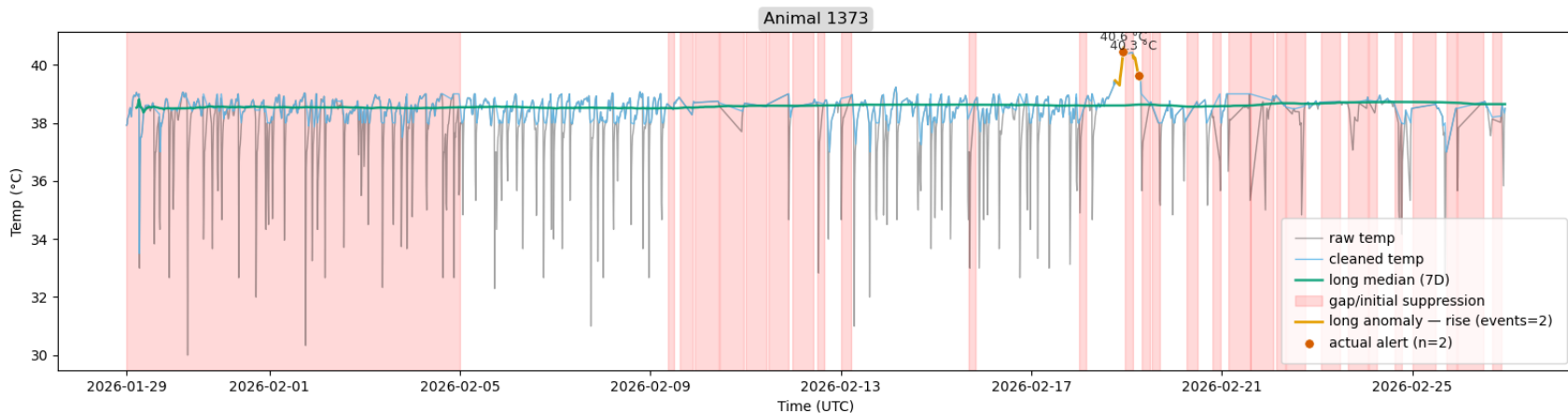
- Detects anomalies using the distribution of deviations (outliers)
- Uses two rolling windows:
 - **Baseline Window:** To get baseline using rolling median and thereby, deviations
 - **Scale Window:** Models historical deviations to define what “normal fluctuation” looks like
- Flags alerts only when:
 - A deviation is unusual relative to past deviations, and
 - It persists over time

Why This Works Better

- Robust to noise and outliers (median + MAD)
- Adapts to each cow's natural variability
- Accounts for seasonality using local context

**Key Shift: “Is this temperature high?” →
“Is this behavior unusually different from
the past—and sustained?”**

Our Method: Context Aware Event Detection



Key Enhancements:

- Group consecutive anomalies into events
- Gap suppression removes false signals from missing data.
- Detect heat stress events earlier than simple quarterly rolling

Challenge: Since temperature drops are highly correlated with water bouts and our method is also an approximation, we do not detect drops since they may be biased.

Model Evaluation Results

Test Set : 30 cows – 20 from dairy and 10 for beef; of those, 50:50 split between cows with alerts and without alerts.

Accuracy: 0.80 | **Precision:** 0.747

Recall: 0.867 | **F1:** 0.813

Earlier Detection

- Match our alerts with Zalliant's by filtering for high alerts and linking only nearby alerts as the same event.
- Median Early Detection = **9 hours** earlier than current alerts!

Model vs Current Alerts

Actual	Predicted	
	detected	not detected
no alerts	4 (26.7%)	11 (73.3%)
alerts	13 (86.7%)	2 (13.3%)

Our model captures **more signals** than current methods and **detects them earlier**, maintaining **robustness to noise** while adjusting for **seasonality** and **individual cow variation**.

Economic Impact: Reduced Sampling Rate

Performance Metrics by Sampling Rate				
Sampling Rate (min)	Precision	Recall	Specificity	Accuracy
5 (baseline)	0.86	0.75	0.92	0.85
10	0.88 (+2%)	0.88 (+17%)	0.92 (+0%)	0.9 (+6%)
15	0.70 (-18%)	0.88 (+17%)	0.75 (-18%)	0.8 (-6%)

Bolus currently records data every 5 minutes

Developed models match or exceed performance with longer sampling windows

Zalliant's Benefit: Improves bolus lifespan and reduces hardware costs by a factor of 2 or 50% (approx.)

Summary

- Achievements & Economic Impact
- Challenges
- Next Steps

Achievements & Economic Impact

- Developed a solution that is interpretable, lightweight, and suitable for large-scale farm implementation under real operational constraints.
- Adjusted sampling frequency (5 → 15 minutes), enabling **~40–50%** reduction in hardware-related lifecycle costs via extended bolus lifespan.
- Enabled ~9-hour earlier identification of heat stress events, supporting earlier and more effective intervention, with estimated ~\$10K annual savings per 100 cows.

Challenges

- Reconstructing baseline temperature to match proprietary methods that have been developed after lots of iterations.
- Constrained Time Length Window: 30 days is the largest time series one can pull
- Sensor Gaps: Large sensor gaps can deviate the baseline which could create false positives
- Our model only detects positive deviations (temperature rises)
- Lack of ground truth alert labels which makes it an unsupervised problem
 - Used Zalliant's current labels as proxy ground truth labels – Assumes Zalliant's alerts are real

Next Steps

- Consolidate final deliverables (poster, report, project website)
- Gather cost metrics to determine economic feasibility and deliver business report

Acknowledgements

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Works Cited

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